

Southeast University Examination Paper (A)

Course Name Principles of Compiling Examination Term 17-18-2 Score _____
Related Major Software Engineering Examination Form Close test Test Duration 180 Mins

There are 8 problems in this paper. You can write the answers in English or Chinese on the attached paper sheets.

1. Please construct context-free grammars with ϵ -free productions for the following language (10%).

$$L = \{\omega \mid \omega \in (a,b)^*, \text{ and } \omega \text{ do not contain the substring } abb\}$$

2. Please construct a DFA with minimum states for the following regular expression. (15%)

$$((ab|ba)(aa|bb)^*(ab|ba)(aa|bb)^*)^*(ab|ba)$$

3. Please eliminate the left recursions (if there are) and extract maximum common left factors (if there are) from the following context free grammar, and then decide the resulted grammar is whether a LL(1) grammar by constructing the related LL(1) parsing table. (15%)

$$S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid a$$

$$E \rightarrow E \text{ and } F \mid F$$

$$F \rightarrow F \text{ or } j|G$$

$$G \rightarrow (E|b$$

4. Please construct a LR(1) parsing table for the following ambiguous grammar with the additional conditions (You determine the required additional conditions by yourself). (15%)

$$S \rightarrow \text{if } E \text{ then } S | \text{if } E \text{ then } S \text{ else } S | a$$

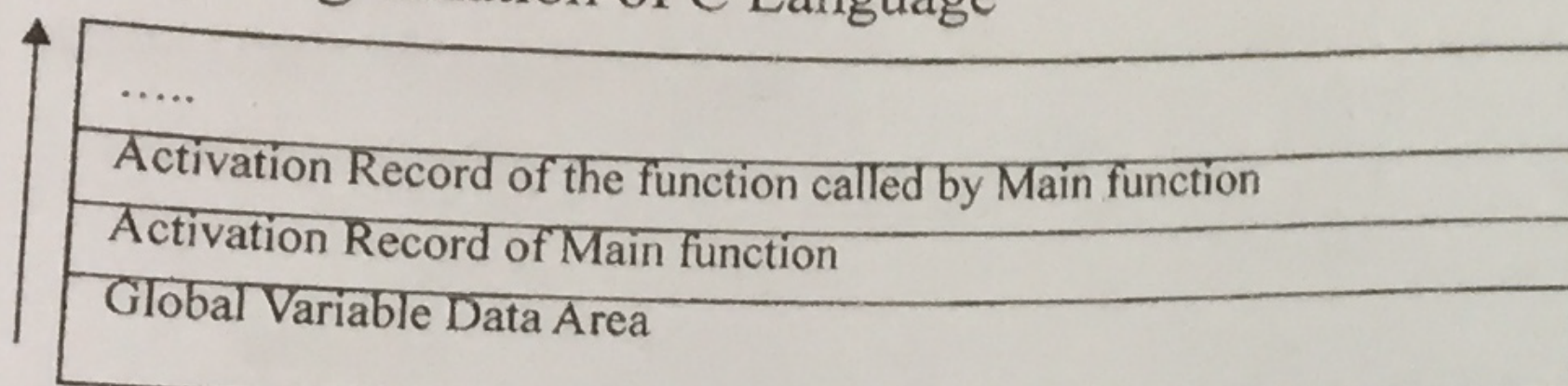
$$E \rightarrow E \text{ and } F | E \text{ or } E | \text{not } E | b$$

5. Please construct an annotated parse tree for the input string 123.321 where the syntax-directed definition is as following (10%):

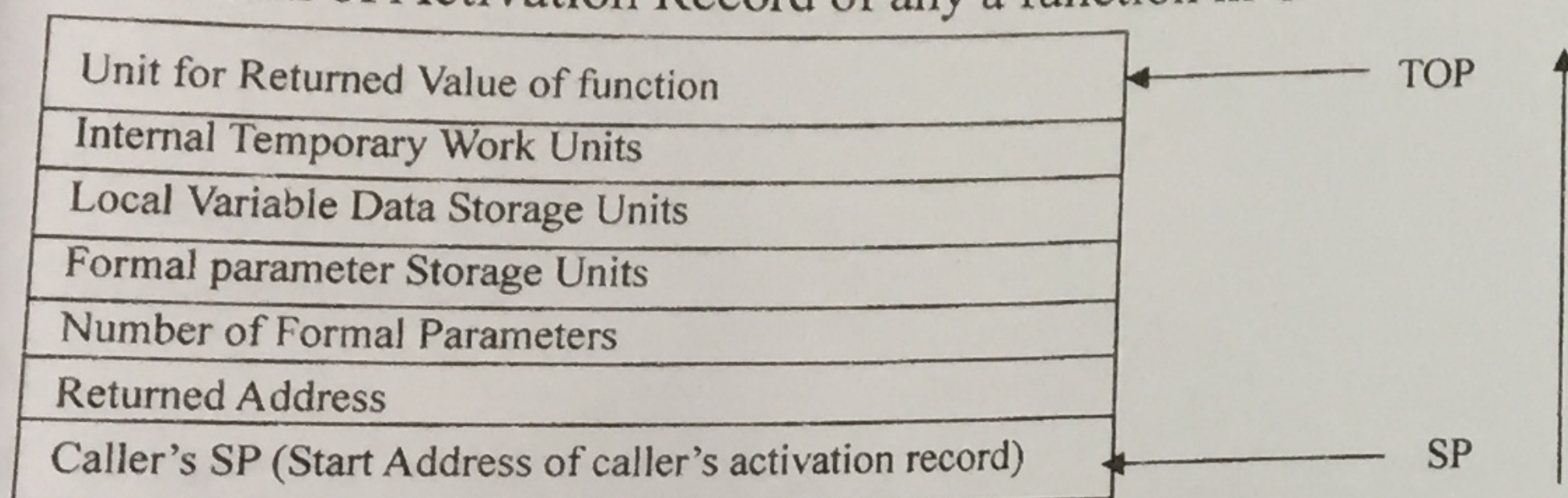
Productions	Semantic Rules
$S \rightarrow L_1.L_2$	$\{S.val = L_1.val + L_2.val / 4^{L_2.len}\}$
$S \rightarrow L$	$\{S.val = L.val\}$
$L \rightarrow L_1Q$	$\{L.val = L_1.val * 4 + Q.val, L.len = L_1.len + 1\}$
$L \rightarrow Q$	$\{L.val = Q.val, L.len = 1\}$
$Q \rightarrow 0$	$\{Q.val = 0\}$
$Q \rightarrow 1$	$\{Q.val = 1\}$
$Q \rightarrow 2$	$\{Q.val = 2\}$
$Q \rightarrow 3$	$\{Q.val = 3\}$

6. We assume that the storage organization and the form of activation record used in C language program run-time stack storage allocation are as following. Please **construct the run-time stack map when it gets the maximum size** for the following C program (10%).

Storage Organization of C Language



The Form of Activation Record of any a function in C



```
#include <stdio.h>
```

```
int x,y,z;
```

```
int main()
```

```
{
    x=9;
    y=5;
    z=C(x,y);
}
```

```
int C(int m,int n)
```

```
{
    if (m>=n)
```



```

if (n<=0) return 1;
else if (n==1) return m;
else {
    int t1;
    t1=m-n;
    if (t1<n)
    {
        int t2;
        t2=C(m,t1);
        return t2
    }
    else {
        int t3,t4,t5,t6,t7;
        t3=m-1;
        t4=n-1;
        t5=C(t3,t4);
        t6=m*t5;
        t7=t6/n;
        return t7;
    }
}
else return 0;
}

```

Notes: 1) Here we assume that the caller's sp of Main function is the start address of global variable data area, and the returned address in the activation record of a function (including Main function) is filled by the operating system automatically, you might not care it.

2) The initial value of variable X and Y are 9 and 5, the start address of stack used in the program is K.

7. Please translate the following program fragment into **quadruple sequence using short circuit code and back-patching techniques.** (15%)

```
i=1;
loop=0;
while (loop==0 && i<=10) {
    j=1;
    while (loop ==0 && j<=20)
        if (a[i,j] != b[i,j])
            {
                loop=1;
                m=i;
                n=j;
            }
        else j=j+1;
    if (loop==0) i=i+1;
};
if (loop==1) e=0;
else e=1;
```

Notes: Here we assume that the declaration of array A and B are array [1..10,1..20], each data element of array A or B would **use 2 storage unit**, and the start address of array A's storage area is addrA, the start address of array B's storage area is addrB.

8. Please **construct the DAG** for the following basic block, and we assume that only variable "M" would be used later, please optimize the block and **rewrite the block** in optimized code form.(10%)

$$B=3$$

$$D=A+C$$

$$E=A*C$$

$$F=D+E$$

$$G=B*F$$

$$H=A+C$$

$$I=A*C$$

$$J=H+I$$

$$K=B*5$$

$$L=K+J$$

$$M=L$$